

CHAPTER 16

Block Designs

In a completely randomized design (CRD), the treatments are assigned to the experimental units at random. This is appropriate when the units are homogeneous, as has been assumed in the designs leading to the one- and two-way analysis of variances (ANOVAs). Sometimes, we may suspect that the units are heterogeneous, but we can not describe the form the difference takes—for example, we may know that a group of patients are not identical, but we may have no further information about them. In this case, it is still appropriate to use a CRD. Of course, the randomization will tend to spread the heterogeneity around to reduce bias, but the real justification lies in the randomization test discussed in Section 3.3. Under the null hypothesis, there is no link between a factor and the response. In other words, the responses have been assigned to the units in a way that is unlinked to the factor. This corresponds to the randomization used in assigning the levels of the factor to the units. This is why the randomization is crucial because it allows us to make this argument. Now if the difference in the response between levels of the factor seems too unlikely to have occurred by chance, we can reject the null hypothesis. The normal-based inference is approximately equivalent to the permutation-based test. Since the normal-based inference is much quicker, we might prefer to use that.

When the experimental units are heterogeneous in a known way and can be arranged into *blocks* where the within block variation is ideally small, but the between block variation is large, a *block design* can be more efficient than a CRD. We prefer to have a block size equal to the number of treatments. If this cannot be done, an *incomplete* block design must be used.

Sometimes the blocks are determined by the experimenter. For example, suppose we want to compare four treatments and have 20 patients available. We might divide the patients into five blocks of four patients each where the patients in each block have some relevant similarity. We might decide this subjectively in the absence of specific information. In other cases, the blocks are predetermined by the nature of the experiment. For example, suppose we want to test three crop varieties on five fields. Restrictions on planting, harvesting and irrigation equipment might allow us only to divide the fields into three strips.

In a randomized block design, the treatment levels are assigned randomly within a block. This means the randomization is restricted relative to the full randomization used in the CRD. This has consequences for the inference. There are fewer possible permutations for the random assignment of the treatments, therefore, the computation of the significance of a statistic based on the permutation test would need to be modified. Similarly, a block effect must be included in the model used for inference about the treatments, even if the block effect is not significant.

16.1 Randomized Block Design

We have one treatment factor, τ at t levels and one blocking factor, β at r levels. The model is:

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$$

where τ_i is the treatment effect and β_j is the blocking effect. There is one observation on each treatment in each block. This is called a randomized complete block design (RCBD). The analysis is then very similar to the two-way ANOVA with one observation per cell. We can check for an interaction and check for a treatment effect. We can also check the block effect, but this is only useful for future reference. Blocking is a feature of the experimental units and restricts the randomized assignment of the treatments. This means that we cannot regain the degrees of freedom devoted to blocking even if the blocking effect is not significant. The randomization test-based argument means that we must judge the magnitude of the treatment effect within the context of the restricted randomization that has been used.

We illustrate this with an experiment to compare eight varieties of oats. The growing area was heterogeneous and so was grouped into five blocks. Each variety was sown once within each block and the yield in grams per 16-ft row was recorded. The data come from Anderson and Bancroft (1952).

We start with a look at the data:

```
> data (oatvar)
> attach (oatvar)
> xtabs (yield ~ variety + block)
      block
variety I   II  III IV   V
      1 296 357 340 331 348
      2 402 390 431 340 320
      3 437 334 426 320 296
      4 303 319 310 260 242
      5 469 405 442 487 394
      6 345 342 358 300 308
      7 324 339 357 352 220
      8 488 374 401 338 320
> stripchart (yield ~ variety, xlab="yield", ylab="variety")
> stripchart (yield ~ block, xlab="yield", ylab="block")
```

See Figure 16.1. There is no indication of outliers, skewness or nonconstant variance. Now check for interactions, as seen in Figure 16.2:

```
> interaction.plot (variety, block, yield)
> interaction.plot (block, variety, yield)
```

There is no clear evidence of interaction:

```
> g <- lm (yield ~ block+variety) >
anova (g)
Analysis of Variance Table
```

Response: yield

Df Sum Sq Mean Sq F value Pr (>F)

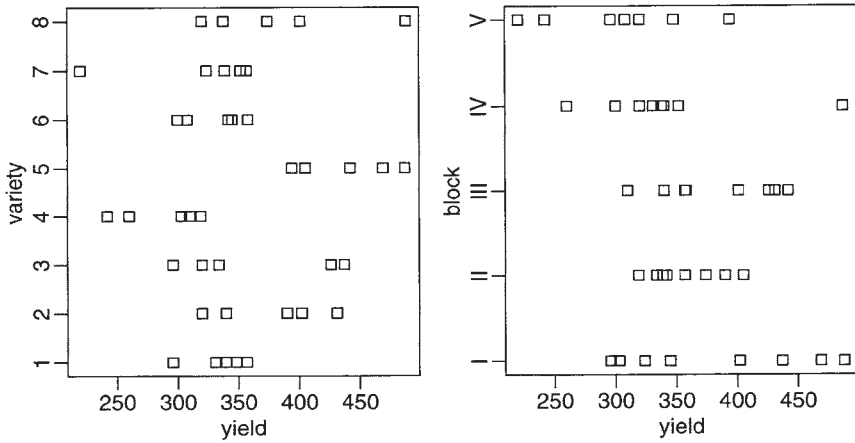


Figure 16.1 Strip plots of oat variety data.

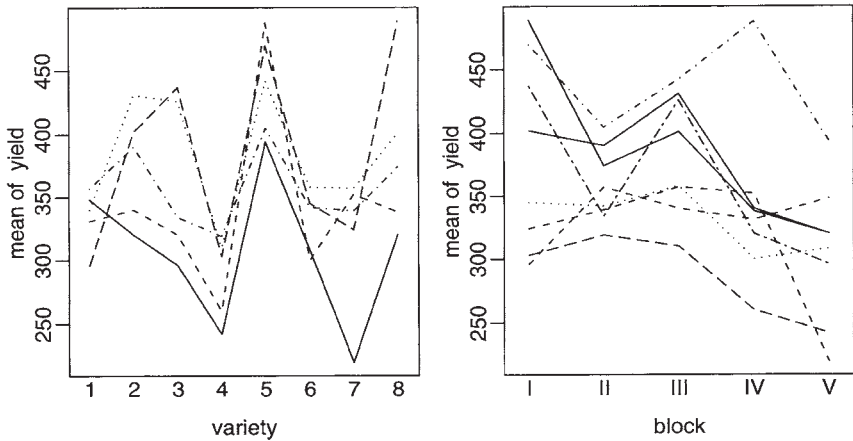


Figure 16.2 Interaction plots for oat variety data.

```

block      4 33396  8349 6.24  0.001
variety    7 77524 11075 8.28  1.8e-05
Residuals 28 37433  1337

```

Both effects are significant. The ANOVA table corresponds to a sequential testing of models, here corresponding to the sequence:

```

y ~ 1
y ~ block
y ~ block+variety

```

So here the p-value 0.001 corresponds to a comparison of the first two models in this list, while the p-value of $1.8\text{e-}05$ corresponds to the test comparing the second two. The denominator in both F-tests is the mean square from the full model, here 1337. This means that a test of the block effect that leaves out the variety effect is not the same:

```
> anova (lm (yield ~ block))
Analysis of Variance Table
Response: yield
      Df Sum Sq Mean Sq F value Pr(>F)
block    4  33396    8349   2.54  0.057
Residuals 35 114957    3284
```

There is a difference in significance in this case. This latter test is incorrect for testing the blocking effect.

Notice that if we change the order of the terms in the ANOVA, it makes no difference because of the orthogonal design:

```
> anova (lm (yield ~ variety+block))
Analysis of Variance Table
Response: yield
      Df Sum Sq Mean Sq F value Pr(>F)
variety  7  77524    11075   8.28 1.8e-05
block    4  33396    8349   6.24  0.001
Residuals 28  37433     1337
```

By way of comparison, see what happens if we omit the first observation in the dataset—this might happen in practice if this run is lost:

```
> anova (lm(yield ~ block+variety,subset=-1))
Analysis of Variance Table
Response: yield
      Df Sum Sq Mean Sq F value Pr(>F)
block    4  38581    9645   8.41 0.00015
variety  7  75339    10763   9.38 7.3e-06
Residuals 27  30968     1147
> anova(lm(yield ~ variety+block, subset=-1))
Analysis of Variance Table
Response: yield
      Df Sum Sq Mean Sq F value Pr(>F)
variety  7  75902    10843   9.45 6.8e-06
block    4  38018     9504   8.29 0.00017
Residuals 27  30968     1147
```

As there is one missing observation, the design is no longer orthogonal and the order does matter, although it would not change the general conclusions. If we want to test for a treatment effect, we would prefer the first of these two tables since in that version the blocking factor is already included when we test the treatment factor. Since the blocking factor is an unalterable feature of the chosen design, this is as it should be. A convenient way to test all terms relative to the full model is:

```
> drop1 (lm (yield ~ variety+block, subset=-1), test="F")
Single term deletions
Model:
yield ~ variety + block
      Df Sum of Sq    RSS AIC F value    Pr(>F)
<none>                 30968 284
variety 7      75339 106307 319     9.38 7.3e-06
block   4       38018  68986 308     8.29 0.00017
```

Check the diagnostics, as seen in Figure 16.3:

```
> plot (fitted (g), residuals (g), xlab="Fitted", ylab="Residuals")
> abline (h=0)
> qqnorm (residuals (g))
> qqline (residuals (g))
```

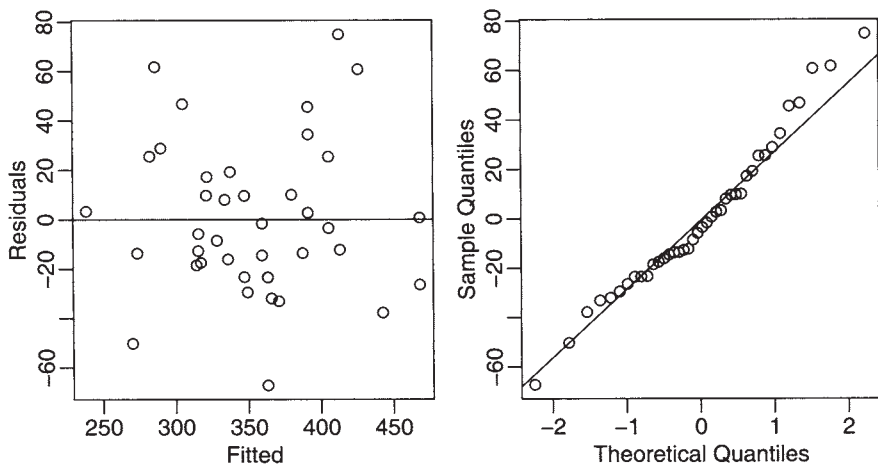


Figure 16.3 Diagnostic plots for the oat variety data.

An examination of which varieties give the highest yields and which are significantly better than others can now follow.

We did assume that the interactions were not significant. We looked at the interaction plots, but we can also execute the Tukey nonadditivity test:

```
> varcoefs <- c (0, coef (g) [6:12])
> blockcoefs <- c (0, coef (g) [2:5])
> ab <- rep (varcoefs, each=5)*rep(blockcoefs,8)
> h <- update (g, .~.+ab)
> anova (h)
Analysis of Variance Table
Response : yield
      Df Sum Sq Mean Sq F value    Pr(>F)
block   4  33396    8349     6.06 0.0013
variety  7  77524   11075     8.03 2.8e-05
```

```
ab          1      213      213      0.15  0.6974
Residuals 27  37220   1379
```

Because the p-value of the treatment times block effect is 0.6974, we accept the null hypothesis of no interaction. Of course, the interaction may be of a nonmultiplicative form, but there is little we can do about that.

Relative advantage of RCBD over CRD

We can measure precision by considering $\text{var } \hat{\tau}$ or equivalently $\hat{\sigma}^2$. We should compare the $\hat{\sigma}^2$ for designs with the same sample size. We define relative efficiency as $\hat{\sigma}_{CRD}^2 / \hat{\sigma}_{RCBD}^2$ where the quantities can be computed by fitting models with and without the blocking effect. For the example above:

```
> gcrd <- lm (yield ~ variety)
> summary (gcrd) $sig
[1] 47.047
> summary (g) $sig
[1] 33.867
> (47.047/33.867)^2
[1] 1.9298
```

So a CRD would require 93% more observations to obtain the same level of precision as an RCBD.

The efficiency is not guaranteed to be greater than one. Only use blocking where there is some heterogeneity in the experimental units. The decision to block is a matter of judgment prior to the experiment. There is no guarantee that it will increase precision.

16.2 Latin Squares

Latin squares are useful when there are two blocking variables. For example, in a field used for agricultural experiments, the level of moisture may vary across the field in one direction and the fertility, in another. In an industrial experiment, suppose we wish to compare four production methods (the treatments)—A, B, C and D. We have available four machines 1, 2, 3 and 4, and four operators, I, II, III and IV. A Latin square design is shown in Table 16.1.

	1	2	3	4
I	A	B	C	D
II	B	D	A	C
III	C	A	D	B
IV	D	C	B	A

Table 16.1 *Latin square showing the treatment (A to D) used for different combinations of two factors.*

Each treatment is assigned to each block once and only once. We should choose randomly from all the possible Latin square layouts.

Let τ be the treatment factor and β and γ be the two blocking factors; then the model is:

$$y_{ijk}=\mu+\tau_i+\beta_j+\gamma_k+\epsilon_{ijk} \quad i, j, k=1,\dots, t$$

All combinations of i, j and k do not appear. To test for a treatment effect simply fit a Vmodel without the treatment effect and compare using the F-test. The Tukey pairwise CIs are:

$$\hat{\tau}_l - \hat{\tau}_m \pm \frac{q_{t,(t-1)(t-2)}}{\sqrt{2}} \hat{\sigma} \sqrt{2/t}$$

The Latin square can be even more efficient than the RCBD provided that the blocking effects are sizable. There are some variations on the Latin square. The Latin square can be replicated if more runs are available. We need to have both block sizes to be equal to the number of treatments. This may be difficult to achieve. Latin rectangle designs are possible by adjoining Latin squares. When there are three blocking variables, a Graeco-Latin square may be used but these rarely arise in practice.

The Latin square can also be used for comparing three treatment factors. Only t^2 runs are required compared to the t^3 required if all combinations were run. (The downside is that you can not estimate the interactions if they exist.) This is an example of a *fractional factorial*.

In an experiment reported by Davies (1954), four materials, A, B, C and D, were fed into a wear-testing machine. The response is the loss of weight in 0.1 mm over the testing period. The machine could process four samples at a time and past experience indicated that there were some differences due to the position of these four samples.

Also some differences were suspected from run to run. Four runs were made:

```
> data(abrasion)
> abrasion
  run position material wear
1    1         1      C  235
```

```
2 1      2      D 236
..etc..
1 4      4      D 225
```

We can check the Latin square structure:

```
> matrix(abrasion$material, 4, 4)
[,1] [,2] [,3] [,4]
[1,] "C" "A" "D" "B"
[2,] "D" "B" "C" "A"
[3,] "B" "D" "A" "C"
[4,] "A" "C" "B" "D"
```

Plot the data:

```
> with (abrasion, stripchart (wear material,
  xlab="Material",vert=T))
> with (abrasion, stripchart (wear run, xlab="Run",vert=T))
> with (abrasion, stripchart (wear position,
  xlab="Position", vert=T))
```

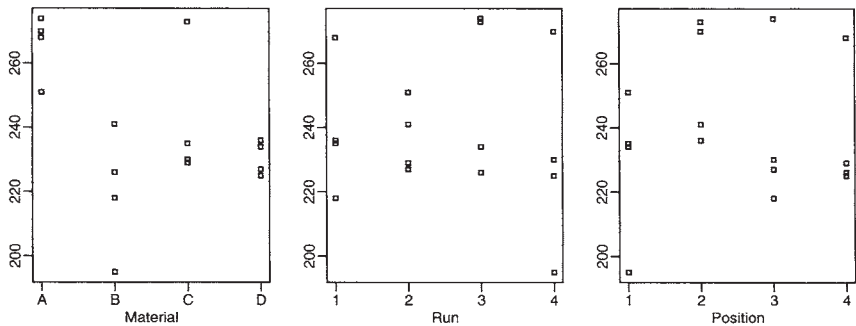


Figure 16.4 *Amount of wear depending on material, run and position.*

Examine the plots in Figure 16.4. There appear to be differences in all variables. No outliers, skewness or unequal variance is apparent.

Now fit the Latin square model and test each variable relative to the full model:

```
> g <- lm (wear ~ material+run+position, abrasion)
> drop1 (g, test="F")
Single term deletions
Model: wear ~ material + run + position
      Df Sum of Sq  RSS   AIC F value    Pr(>F)
<none>                 368    70
material 3         4621 4989   106   25.15 0.00085
run       3         986 1354    85    5.37 0.03901
position 3        1468 1836    90    7.99 0.01617
```

We see that all variables are statistically significant. There are clear differences between the materials. We checked the diagnostics on the model, which showed nothing remarkable. We examine the coefficients:

```
> summary (g)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   254.75      6.19    41.17 14e-08
materialB     -45.75      5.53    -8.27 0.00017
materialC     -24.00      5.53    -4.34 0.00489
materialD     -35.25      5.53    -6.37 0.00070
run2           -2.25      5.53    -0.41 0.69842
run3           12.50      5.53     2.26 0.06466
run4           -9.25      5.53    -1.67 0.14566
position2      26.25      5.53     4.74 0.00318
position3       8.50      5.53     1.54 0.17545
position4       8.25      5.53     1.49 0.18661
Residual standard error: 7.83 on 6 degrees of freedom
Multiple R-Squared: 0.951, Adjusted R-squared: 0.877
F-statistic: 12.8 on 9 and 6 DF, p-value: 0.00283
```

We see that material B looks best (in terms of least wear) followed by material D. Is the difference significant though? Which materials in general are significantly better than others? We need the Tukey pairwise intervals to help determine this. The width of the band is calculated in the usual manner:

```
> qtukey (0.95, 4, 6)*5.53/sqrt(2)
[1] 19.143
```

The width of the interval is 19.1. We can make a table of the material differences:

```
> scoefs <- c (0, coef (g) [2:4])
> outer (scoefs, scoefs, "-")
              materialB materialC materialD
0.00          45.75      24.00      35.25
materialB -45.75          0.00-21.75     -10.50
materialC -24.00      21.75          0.00      11.25
materialD -35.25      10.50     -11.25          0.00
```

We see that the (B, D) and (D, C) differences are not significant at the 5% level, but that all the other differences are significant.

If maximizing resistance to wear is our aim, we would pick material B but if material D offered a better price, we might have some cause to consider switching to D. The decision would need to be made with cost-quality trade-offs in mind.

Now we compute how efficient the Latin square is compared to other designs. We compare to the completely randomized design:

```
> gr <- lm (wear ~ material, abrasion)
> (summary (gr) $sig/summary (g) $sig) ~ 2
[1] 3.8401
```

We see that the Latin square is 3.84 times more efficient than the CRD. This is a substantial gain in efficiency. The Latin square may also be combined to designs where we block on only one of the variables. The efficiency relative to these designs is less impressive, but still worthwhile.

16.3 Balanced Incomplete Block Design

In a complete block design, the block size is equal to the number of treatments. When the block size is less than the number of treatments, an incomplete block design must be used. For example, in the oat example, suppose six oat varieties were to be compared, but each field had space for only four plots.

In an incomplete block design, the treatments and blocks are *not* orthogonal. Some treatment contrasts will not be identifiable from certain block contrasts. This is an example of *confounding*. This means that those treatment contrasts effectively cannot be examined. In a balanced incomplete block (BIB) design, all the pairwise differences are identifiable and have the same standard error. Pairwise differences are more likely to be interesting than other contrasts, so the design is constructed to facilitate this.

Suppose, we have four treatments ($t=4$) A, B, C, D and the block size, $k=3$ and there are $b=4$ blocks. Therefore, each treatment appears $r=3$ times in the design. One possible BIB design is:

Block 1	A	B	C
Block 2	A	B	D
Block 3	A	C	D
Block 4	B	C	D

Table 16.2 *BIB design for four treatments with four blocks of size three.*

Each pair of treatments appears in the same block $\lambda=2$ times—this feature means simpler pairwise comparison is possible. For a BIB design, we require:

$$\begin{aligned} b &\geq t > k \\ rt &= bk = n \\ \lambda(t-1) &= r(k-1) \end{aligned}$$

This last relation holds because the number of pairs in a block is $k(k-1)/2$ so the total number of pairs must be $bk(k-1)/2$. On the other hand, the number of treatment pairs is $t(t-1)/2$. The ratio of these two quantities must be λ .

Since λ has to be an integer, a BIB design is not always possible even when the first two conditions are satisfied. For example, consider $r=4$, $t=3$, $b=6$, $k=2$ and then $\lambda=2$ which is OK, but if $r=4$, $t=4$, $b=8$, $k=2$, then $\lambda=4/3$ so no BIB is possible. (Something called a partially balanced incomplete block design can then be used.) BIBs are also useful for competitions where not all contestants can fit in the same race.

The model we fit is the same as for the RCBD:

$$y_{ij} = \mu + \tau_i + \beta_j + \epsilon_{ij}$$

In our example, a nutritionist studied the effects of six diets, “a” through “f,” on weight gain of domestic rabbits. From past experience with sizes of litters, it was felt that only three uniform rabbits could be selected from each available litter. There were ten litters available forming blocks of size three. The data come from Lentner and Bishop (1986). Examine the data:

```
> data(rabbit)
> xtabs (gain ~ treat+block, rabbit)
block
treat  b1  b10  b2  b3  b4  b5  b6  b7  b8  b9
a      0.0 37.3 40.1 0.0 44.9 0.0 45.2 44.0 0.0
b      32.6 0.0 38.1 0.0 0.0 0.0 37.3 40.6 0.0 30.6
c      35.2 0.0 40.9 34.6 43.9 40.9 0.0 0.0 0.0 0.0
d      0.0 42.3 0.0 37.5 0.0 37.3 0.0 37.9 0.0 27.5
e      0.0 0.0 0.0 0.0 40.8 32.0 40.5 0.0 38.5 20.6
f      42.2 41.7 0.0 34.3 0.0 0.0 42.8 0.0 51.9 0.0
```

The zero values correspond to no observation. The BIB structure is apparent—each pair of diets appear in the same block exactly twice. Now plot the data, as seen in Figure 16.5:

```
> attach(rabbit)
> stripchart (gain ~ block, xlab="weight gain", ylab="block")
> stripchart (gain ~ treat, xlab="weight gain", ylab="treat")
```

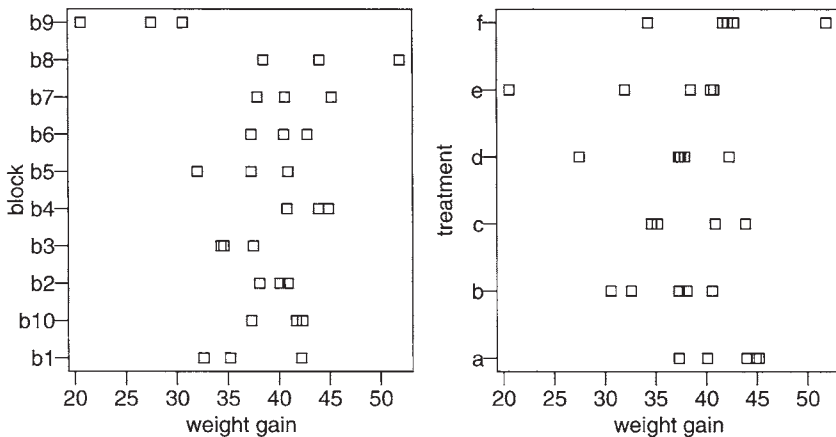


Figure 16.5 Strip plots of rabbit diet data.

We fit the model taking care to put the blocking factor first:

```
> g <- lm (gain ~ block+treat,rabbit) >
anova (g)
Analysis of Variance Table
Response: gain
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
block	9	730	81	8.07	0.00025
treat	5	159	32	3.16	0.03817
Residuals	15	151	10		

Changing the order of treatment and block:

```
Analysis of Variance Table
> anova (lm(gain ~ treat+block, rabbit))
Response: gain
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
treat	5	293	59	5.84	0.00345
block	9	596	66	6.59	0.00076
Residuals	15	151	10		

does make a difference because the design is not orthogonal because of the incompleteness. We prefer the first table, because we want to test for a treatment effect after the blocking effect has been considered.

Now check the diagnostics:

```
> plot (fitted (g), residuals (g), xlab="Fittedn, ylab="Residuals")
> abline (h=0)
> qqnorm (residuals (g), main="")
> qqline (residuals (g))
```

The plots are not shown, as they show nothing of interest.

Now we check which treatments differ. The Tukey pairwise CIs need to be directly constructed because this is not a complete layout. We extract the information about the treatment effects and the standard error:

```
> coef (summary (g))
```

	Estimate	Std. Error	t value	Pr(> t)
treatb	-1.741667	2.2418	-0.776898	0.449298
treatc	0.400000	2.2418	0.178426	0.860776
treatd	0.066667	2.2418	0.029738	0.976668
treate	-5.225000	2.2418	-2.330695	0.034134
treatf	3.300000	2.2418	1.472018	0.161686

We see that the standard error for the pairwise comparison is 2.24. Notice that all the treatment standard errors are equal because of the BIB. Now compute the Tukey critical value:

```
> qtukey(0.95, 6, 15)
[1] 4.5947
```

So the intervals have width:

```
> 4.59*2.24/sqrt (2)
[1] 7.2702
```

We check which pairs are significantly different:

```
> tcoefs <- c (0, coef (g) [11:15])
> abs (outer (tcoefs, tcoefs, "-")) > 7.27
      treatb treatc treatd treate treatf
treatb FALSE  FALSE  FALSE  FALSE  FALSE  FALSE
treatc FALSE  FALSE  FALSE  FALSE  FALSE  FALSE
treatd FALSE  FALSE  FALSE  FALSE  FALSE  FALSE
treate FALSE  FALSE  FALSE  FALSE  FALSE  TRUE
treatf FALSE  FALSE  FALSE  FALSE   TRUE  FALSE
```

Only the e - f difference is significant.

Now let's see how much better this blocked design is than the CRD. We compute the relative efficiency:

```
> gr <- lm (gain ~ treat, rabbit)
> (summary (gr) $sig/summary (g) $sig)^2
[1] 3.0945
```

Blocking was well worthwhile here.

Exercises

1. The alf alf data arise from a Latin square design where the treatment factor is inoculum and the blocking factors are shade and irrigation. Test the significance of the effects and determine which levels of the treatment factor are significantly different.
2. The eggprod comes from a randomized block experiment to determine factors affecting egg production. Is there a difference in the treatments? What is the nature of the difference? What efficiency was gained by the blocked design?
3. The morley data can be viewed as a randomized block experiment with Run as the treatment factor and Expt as the blocking factor. Is there a difference between runs and what efficiency is gained by blocking?
4. The OrchardSprays data arise from a Latin square design. Which factors show a difference? What pairs of treatments are different? Can the treatment factor be replaced with a linear term? What efficiency is gained by this design?
5. The resceram data arise from an experiment to test the effects of resistor shape on current noise. The resistors are mounted on plates and only three resistors will fit on a plate although there are four different shapes to be tested. Identify the design and analyze. Which shapes are different?

